

KAHLER ASTEN, Germany

WMO: 104270

Lat: 51.1803N

Lon: 8.4892E

Elev: 2817

StdP: 13.26

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.6	11.0	-3.5	5.1	12.4	2.2	6.8	13.2	35.5	38.7	32.0	30.7	14.6	90	0.804

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	74.7	61.4	71.7	59.6	68.8	58.4	62.8	71.3	61.2	69.1	59.6	66.6	10.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.7	84.8	65.5	58.1	80.2	63.7	56.7	76.2	62.2	29.8	71.1	28.6	69.4	27.5	66.7	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.1	27.1	24.6	5.3	80.7	8.5	2.9	-0.8	82.7	-5.8	84.4	-10.5	86.0	-16.6	88.1
			4.5	66.5	8.3	1.5	-1.5	67.6	-6.4	68.5	-11.0	69.4	-17.1	70.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	27.9	27.1	33.2	38.5	47.9	52.5	56.8	56.7	50.7	43.8	35.0	31.3
	DBStd	12.77	8.39	8.92	7.29	7.38	7.67	7.39	6.76	6.45	6.20	6.51	7.25	6.23
	HDD50	3725	687	642	522	349	137	57	13	12	65	212	450	579
	HDD65	8469	1152	1062	986	794	531	379	267	268	430	656	900	1044
	CDD50	759	0	0	1	5	71	131	224	220	85	21	1	0
	CDD65	27	0	0	0	0	0	4	13	10	0	0	0	0
	CDH74	114	0	0	0	0	0	10	53	51	0	0	0	0
	CDH80	8	0	0	0	0	0	0	1	7	0	0	0	0
	WSAvg	13.8	16.8	15.9	14.7	13.0	11.8	11.5	11.6	11.4	12.8	14.5	15.3	16.0
	PrecAvg	43.5	4.5	3.6	3.3	2.5	3.1	3.3	3.7	3.6	3.4	3.5	4.2	4.7
(15) Precipitation	PrecMax	58.7	10.0	8.7	7.5	4.6	6.9	5.4	6.3	8.0	8.3	12.1	7.4	11.3
	PrecMin	33.6	0.2	0.9	0.5	0.0	1.3	1.7	1.6	0.8	0.9	0.9	0.1	1.1
	PrecStd	6.7	2.4	2.0	1.9	1.2	1.4	1.0	1.3	1.8	1.8	2.1	1.8	2.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	49.8	57.0	62.3	71.9	75.4	78.5	79.9	72.1	64.7	54.8	47.7
		MCWB	38.0	41.8	45.7	49.5	56.3	61.9	62.4	62.6	58.2	55.4	45.3	43.5
	2%	DB	42.3	45.4	50.8	57.1	67.4	71.3	75.2	74.0	67.6	59.0	50.8	44.6
		MCWB	38.2	40.9	42.8	46.6	53.4	59.6	62.0	61.0	57.6	52.3	44.5	42.3
	5%	DB	39.9	41.6	46.1	53.6	63.8	68.0	71.9	70.6	63.2	55.8	47.3	42.3
		MCWB	37.6	38.3	41.6	44.6	52.2	57.8	60.2	59.7	55.9	51.1	44.3	40.1
	10%	DB	37.5	38.2	42.5	49.7	60.4	64.4	68.4	67.2	59.6	52.5	44.9	39.9
		MCWB	36.0	36.2	39.7	42.8	50.8	55.8	58.8	58.2	54.3	49.7	42.6	38.3
	0.4%	WB	42.9	45.0	46.7	50.5	57.4	64.2	65.2	64.7	61.8	57.0	49.0	45.6
		MCDB	43.1	47.6	52.5	59.0	66.4	71.8	74.0	76.4	67.5	61.2	50.7	46.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.4	41.8	44.2	48.1	55.5	61.2	63.0	62.1	59.0	54.4	46.6	43.1
		MCDB	40.6	44.3	48.2	54.5	63.9	68.7	71.9	70.7	64.4	57.4	48.1	43.9
	5%	WB	38.2	38.6	42.2	45.8	53.5	59.0	61.4	60.6	57.0	52.1	44.9	41.0
		MCDB	38.8	40.8	45.6	51.6	62.3	65.9	69.7	68.4	60.8	54.2	46.8	41.5
	10%	WB	36.0	36.3	40.0	43.6	51.6	56.8	59.7	59.1	55.3	50.0	42.8	39.0
		MCDB	36.8	38.0	42.7	48.8	59.2	63.3	67.0	65.3	58.9	51.9	44.4	39.5
	5% DB	MDBR	7.0	8.3	8.7	11.5	13.5	12.5	13.5	12.9	10.8	8.9	7.1	6.5
		MCDBR	10.1	12.6	13.4	16.2	18.6	17.1	18.1	17.4	15.7	13.0	11.1	9.3
		MCWBR	8.6	8.8	8.0	8.6	9.6	9.1	8.9	8.6	8.7	8.0	8.4	7.7
		MCWBR	8.5	11.3	11.7	15.2	16.5	15.6	16.8	15.6	13.5	10.2	9.9	7.6
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.8	8.8	8.1	9.1	9.3	9.3	9.2	8.5	8.2	7.5	8.3	7.5
	taub		0.282	0.304	0.333	0.362	0.366	0.367	0.374	0.370	0.343	0.323	0.303	0.281
		taud	2.456	2.422	2.379	2.339	2.347	2.355	2.326	2.363	2.436	2.487	2.474	2.448
(40) Clear-Sky Solar Irradiance	Ebn at Noon		241	264	275	278	282	282	277	272	269	253	229	221
		Edh at Noon	18	24	31	36	37	37	38	35	29	23	18	16
	RadAvg		226	442	794	1269	1518	1663	1577	1350	971	568	255	168
(44) All-Sky Solar Radiation	RadStd		33	70	109	158	164	154	206	126	134	59	37	20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page